



Carbon black

Carbon black (with subtypes **acetylene black**, **channel black**, **furnace black**, **lamp black** and **thermal black**) is a material produced by the incomplete combustion of coal tar, vegetable matter, or petroleum products, including fuel oil, fluid catalytic cracking tar, and ethylene cracking in a limited supply of air. Carbon black is a form of paracrystalline carbon that has a high surface-area-to-volume ratio, albeit lower than that of activated carbon. It is dissimilar to soot in its much higher surface-area-to-volume ratio and significantly lower (negligible and non-bioavailable) polycyclic aromatic hydrocarbon (PAH) content.

Carbon black is used as a colorant and reinforcing filler in tires and other rubber products and as a pigment and wear protection additive in plastics, paints, and ink pigment.^[2] It is used in the EU as a food colorant when produced from vegetable matter (E153).

The current International Agency for Research on Cancer (IARC) evaluation is that, "Carbon black is possibly carcinogenic to humans (Group 2B)".^[3] Short-term exposure to high concentrations of carbon black dust may produce discomfort to the upper respiratory tract through mechanical irritation.

Common uses

The most common use (70%) of carbon black is as a pigment and reinforcing phase in automobile tires. Carbon black also helps conduct heat away from the tread and belt area of the tire, reducing thermal damage and increasing tire life. Its low cost makes it a common addition to cathodes and anodes and is considered a safe replacement to lithium metal in lithium-ion batteries.^[4] About 20% of world production goes into belts, hoses, and other non-tire rubber goods. The remaining 10% use of carbon black comes from

Carbon black



Names	
Other names	Acetylene black Channel black Furnace black Lamp black Thermal black C.I. Pigment Black 6
Identifiers	
CAS Number	1333-86-4 (https://commonchemistry.cas.org/detail?cas_rn=1333-86-4) ✓
3D model (JSmol)	Interactive image (https://chemapps.stolaf.edu/jmol/jmol.php?model=C)
ECHA InfoCard	100.014.191 (https://echa.europa.eu/substance-information/-/substanceinfo/100.014.191)
EC Number	215-609-9
E number	E152 (colours)
UNII	4XYU5U00C4 (https://precision.fda.gov/uniisearch/srs/uni/4XYU5U00C4) ✓
CompTox Dashboard (EPA)	DTXSID7051216 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID7051216)
SMILES	

pigment in inks, coatings, and plastics, as well as being used as a conductive additive in lithium-ion batteries.^[5]

Carbon black is added to polypropylene because it absorbs ultraviolet radiation, which otherwise causes the material to degrade. Carbon black particles are also employed in some radar absorbent materials, in photocopier and laser printer toner, and in other inks and paints. The high tinting strength and stability of carbon black has also provided use in coloring of resins and films.^[6] Carbon black has been used in various applications for electronics. A good conductor of electricity, carbon black is used as a filler mixed in plastics, elastomer, films, adhesives, and paints.^[6] It is used as an antistatic additive agent in automobile fuel caps and pipes.

Carbon black from vegetable origin is used as a food coloring, known in Europe as additive **E153**. It is approved for use as additive **153** (Carbon blacks or Vegetable carbon) in Australia and New Zealand^[7] but has been banned in the US.^[8] The color pigment carbon black has been widely used for many years in food and beverage packaging. It is used in multi-layer UHT milk bottles in the US, parts of Europe and Asia, and South Africa, and in items like microwavable meal trays and meat trays in New Zealand.

The Canadian Government's extensive review of carbon black in 2011 concluded that carbon black could continue to be used in products – including food packaging for consumers – in Canada. This was because "in most consumer products carbon black is bound in a matrix and unavailable for exposure, for example as a pigment in plastics and rubbers" and "it is proposed that carbon black is not entering the environment in a quantity or concentrations or under conditions that constitute or may constitute a danger in Canada to human life or health."^[9]

Within Australasia, the color pigment carbon black in packaging must comply with the requirements of either the EU or US packaging regulations. If any colorant is used, it must meet European partial agreement AP(89)1.^[10]

Total production was around 8,100,000 metric tons (8,900,000 short tons) in 2006.^[11] Global consumption of carbon black, estimated at 13.2 million metric tons, valued at US\$13.7 billion, in 2015, is expected to reach 13.9 million metric tons, valued at US\$14.4 billion in 2016.

C	
Properties	
Chemical formula	C
Molar mass	12.011 g·mol ^{−1}
Appearance	Black solid
Density	1.8–2.1 g/cm ³ (20 °C) ^[1]
Solubility in water	Practically insoluble ^[1]
Hazards	
Lethal dose or concentration (LD, LC):	
LD ₅₀ (median dose)	> 15400 mg/kg (oral rat) ^[1] 3000 mg/kg (dermal, rabbit) ^[1]
Except where otherwise noted, data are given for materials in their <u>standard state</u> (at 25 °C [77 °F], 100 kPa).	
<u>Infobox references</u>	



Worker at carbon black plant, 1942

While distinct from soot and similar particulates, carbon black can be used as a model compound for diesel soot to better understand how diesel soot behaves under various reaction conditions. Carbon black and diesel soot have some similar properties such as particle sizes, densities, and copolymer adsorption abilities that contribute to them having similar behaviours under various reactions such as oxidation experiments.^{[12][13]}

Global consumption is forecast to maintain a CAGR (compound annual growth rate) of 5.6% between 2016 and 2022, reaching 19.2 million metric tons, valued at US\$20.4 billion, by 2022.^[14]

Reinforcing carbon blacks

The highest volume use of carbon black is as a reinforcing filler in rubber products, especially tires. While a pure gum vulcanization of styrene-butadiene has a tensile strength of no more than 2 MPa and negligible abrasion resistance, compounding it with 50% carbon black by weight improves its tensile strength and wear resistance as shown in the table below. It is used often in the aerospace industry in elastomers for aircraft vibration control components such as engine mounts.

Certain types of carbon black used in tires, plastics and paints

Name	Abbrev.	ASTM desig.	Particle Size nm	Tensile strength MPa	Relative laboratory abrasion	Relative roadwear abrasion
Super Abrasion Furnace	SAF	N110	20–25	25.2	1.35	1.25
Intermediate SAF	ISAF	N220	24–33	23.1	1.25	1.15
High Abrasion Furnace	HAF	N330	28–36	22.4	1.00	1.00
Easy Processing Channel	EPC	N300	30–35	21.7	0.80	0.90
Fast Extruding Furnace	FEF	N550	39–55	18.2	0.64	0.72
High Modulus Furnace	HMF	N660	49–73	16.1	0.56	0.66
Semi-Reinforcing Furnace	SRF	N770	70–96	14.7	0.48	0.60
Fine Thermal	FT	N880	180–200	12.6	0.22	–
Medium Thermal	MT	N990	250–350	9.8	0.18	–

Practically all rubber products where tensile and abrasion wear properties are important use carbon black, so they are black in color. Where physical properties are important but colors other than black are desired, such as white tennis shoes, precipitated or fumed silica has been substituted for carbon black. Silica-based fillers are also gaining market share in automotive tires because they provide better trade-off for fuel efficiency and wet handling due to a lower rolling loss. Traditionally silica fillers had worse abrasion wear properties, but the technology has gradually improved to a point where they can match carbon black abrasion performance.

Pigment

Carbon black (Color Index International, PBK-7) is the name of a common black pigment, traditionally produced from charring organic materials such as wood or bone. It appears black because it reflects very little light in the visible part of the spectrum, with an albedo near zero. The actual albedo varies depending on the source material and method of production. It is known by a variety of names, each of which reflects a traditional method for producing carbon black:

- **Ivory black** was traditionally produced by charring ivory or bones (see bone char).
- **Vine black** was traditionally produced by charring desiccated grape vines and stems.
- **Lamp black** was traditionally produced by collecting soot from oil lamps.

All of these types of carbon black were used extensively as paint pigments since prehistoric times.^[15] Rembrandt, Vermeer, Van Dyck, and more recently, Cézanne, Picasso and Manet^[16] employed carbon black pigments in their paintings. A typical example is Manet's "Music in the Tuileries",^[17] where the black dresses and the men's hats are painted in ivory black.^[18]

Newer methods of producing carbon black have largely superseded these traditional sources. For artisanal purposes, carbon black produced by any means remains common.^[6]

Surface and surface chemistry

All carbon blacks have chemisorbed oxygen complexes (i.e., carboxylic, quinonic, lactonic, phenolic groups and others) on their surfaces to varying degrees depending on the conditions of manufacture.^[19] These surface oxygen groups are collectively referred to as volatile content. It is also known to be a non-conductive material due to its volatile content.

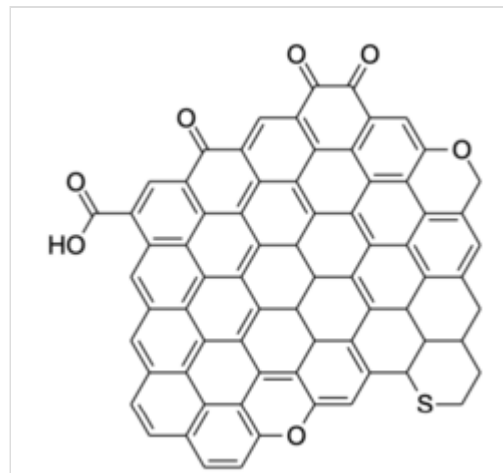
The coatings and inks industries prefer grades of carbon black that are acid-oxidized. Acid is sprayed in high-temperature dryers during the manufacturing process to change the inherent surface chemistry of the black. The amount of chemically-bonded oxygen on the surface area of the black is increased to enhance performance characteristics.

Use in lithium-ion batteries

Carbon black is a common conductive additive for lithium-ion batteries as the particles have small sizes and a large specific surface areas (SSA) which allow for the additive to be well distributed throughout the cathode or anode in addition to being cheap and long-lasting.^{[5][20]} Unlike graphite, which is one of the other common materials used in chargeable batteries, carbon black consists of crystal lattices that are further apart and promotes Li^+ intercalation because it allows more pathways for lithium storage.^[20]

Carbon black has a low density that allows for a large volume of it to be dispersed so that its conductive effects are applied evenly throughout the battery.^{[21][22]} Furthermore, its arrangement of randomly distributed graphite-like crystals improves battery stability because of the decrease in the potential barrier of lithium intercalation into graphite, which ultimately affects the performance of cathodes.^[20]

While carbon black is lightweight and well dispersed throughout the battery and increases the conductive performance of batteries, it also contains oxygen containing hydrophilic functional groups that can cause side reactions to occur in the battery and lead to the decomposition of electrolyte. Graphitization (heating) of carbon black can thermally decompose the hydrophilic functional groups and thus increase the cycle life of the battery which maintains the conductive abilities of carbon black while mitigating the damage that can be caused to batteries by hydrophilic functional groups.



The generic structure of carbon black

Half cells created with heavy graphitization, light graphitization, and no graphitization showed that the cell created with heavy graphitization had a stable cycle life of 320 cycles, the cell with light graphitization showed a stable cycle life of 200 cycles, and the cell with no graphitization showed a stable cycle life of 160 cycles.^[5]

Safety

Carcinogenicity

Carbon black is considered *possibly carcinogenic to humans* and classified as a Group 2B carcinogen because there is sufficient evidence in experimental animals with inadequate evidence in human epidemiological studies.^[3] The evidence of carcinogenicity in animal studies comes from two chronic inhalation studies and two intratracheal instillation studies in rats, which showed significantly elevated rates of lung cancer in exposed animals.^[3] An inhalation study on mice did not show significantly elevated rates of lung cancer in exposed animals.^[3] Epidemiologic data comes from three cohort studies of carbon black production workers. Two studies, from the United Kingdom and Germany, with over 1,000 workers in each study group showed elevated mortality from lung cancer.^[3] A third study of over 5,000 carbon black workers in the United States did not show elevated mortality.^[3] Newer findings of increased lung cancer mortality in an update from the UK study suggest that carbon black could be a late-stage carcinogen.^{[23][24]} However, a more recent and larger study from Germany did not confirm this hypothesis.^[25]

Occupational safety

There are strict guidelines available and in place to ensure employees who manufacture carbon black are not at risk of inhaling unsafe doses of carbon black in its raw form.^[26] Respiratory personal protective equipment is recommended to properly protect workers from inhalation of carbon black. The recommended type of respiratory protection varies depending on the concentration of carbon black used.^[27]

People can be exposed to carbon black in the workplace by inhalation and contact with the skin or eyes. The Occupational Safety and Health Administration (OSHA) has set the legal limit (Permissible exposure limit) for carbon black exposure in the workplace at 3.5 mg/m³ over an 8-hour workday. The National Institute for Occupational Safety and Health (NIOSH) has set a Recommended exposure limit (REL) of 3.5 mg/m³ over an 8-hour workday. At levels of 1750 mg/m³, carbon black is immediately dangerous to life and health.^[28]

See also

- Activated carbon
- Biochar
- Electroconductive carbon black
- George Oenslager - early use of carbon black in rubber
- Heavy metals
- Jean-Baptiste Donnet - carbon black pioneer
- Joseph C. Krejci - carbon black pioneer
- Kværner process - production from hydrocarbons
- List of inorganic pigments
- Siegfried Wolff - carbon black pioneer
- William B. Wiegand - carbon black pioneer

References

1. Record (<https://gestis.dguv.de/data?name=091940&lang=en>) in the GESTIS Substance Database of the Institute for Occupational Safety and Health
2. "Market Study: Carbon Black" (<http://www.ceresana.com/en/market-studies/chemicals/carbon-black/>). Ceresana. Retrieved 2013-04-26.
3. Kuempel, Eileen D.; Sorahan, Tom (2010). "Identification of Research Needs to Resolve the Carcinogenicity of High-priority IARC Carcinogens" (<https://web.archive.org/web/20160304052329/http://monographs.iarc.fr/ENG/Publications/techrep42/TR42-Full.pdf>) (PDF). *Views and Expert Opinions of an IARC/NORA Expert Group Meeting, Lyon, France, 30 June – 2 July 2009. IARC Technical Publication No. 42. Lyon, France: International Agency for Research on Cancer.* **42**: 61–72. Archived from the original (<http://monographs.iarc.fr/ENG/Publications/techrep42/TR42-Full.pdf>) (PDF) on 4 March 2016. Retrieved August 30, 2012.
4. Gnanamuthu, RM.; Lee, Chang Woo (2011-11-01). "Electrochemical properties of Super P carbon black as an anode active material for lithium-ion batteries" (<https://www.sciencedirect.com/science/article/pii/S0254058411007498>). *Materials Chemistry and Physics.* **130** (3): 831–834. doi:10.1016/j.matchemphys.2011.08.060 (<https://doi.org/10.1016%2Fj.matchemphys.2011.08.060>). ISSN 0254-0584 (<https://search.worldcat.org/issn/0254-0584>).
5. Qi, Xin; Blizanac, Berislav; DuPasquier, Aurelien; Lal, Archit; Niehoff, Philip; Placke, Tobias; Oljaca, Miodrag; Li, Jie; Winter, Martin (2015). "Influence of Thermal Treated Carbon Black Conductive Additive on the Performance of High Voltage Spinel Cr-Doped LiNi 0.5 Mn 1.5 O 4 Composite Cathode Electrode" (<https://iopscience.iop.org/article/10.1149/2.0401503jes>). *Journal of the Electrochemical Society.* **162** (3): A339 – A343. doi:10.1149/2.0401503jes (<https://doi.org/10.1149%2F2.0401503jes>). ISSN 0013-4651 (<https://search.worldcat.org/issn/0013-4651>).
6. "Application Examples of carbon black" (<http://www.carbonblack.jp/en/cb/youto.html>). Mitsubishi Chemical. Retrieved 2013-01-14.

7. Australia New Zealand Food Standards Code "Standard 1.2.4 – Labelling of ingredients" (<http://www.comlaw.gov.au/Details/F2011C00827>). 8 September 2011. Retrieved 2011-10-27.
8. US FDA: "Color Additive Status List" (<https://web.archive.org/web/20090606162201/http://www.fda.gov/ForIndustry/ColorAdditives/ColorAdditiveInventories/ucm106626.htm>). *Food and Drug Administration*. Archived from the original (<https://www.fda.gov/ForIndustry/ColorAdditives/ColorAdditiveInventories/ucm106626.htm>) on June 6, 2009. Retrieved 2011-10-27.
9. "Draft Screening Assessment for the Challenge" (<http://www.ec.gc.ca/ese-ees/default.asp?lang=en&n=2cf34283-1>). 29 January 2010. Retrieved 2013-01-14.
10. "Australia New Zealand Food Standards Code" (<https://web.archive.org/web/20121220061636/http://www.foodstandards.govt.nz/foodstandards/foodstandardscode.cfm>). Archived from the original (<http://www.foodstandards.govt.nz/foodstandards/foodstandardscode.cfm>) on 2012-12-20. Retrieved 2013-01-14.
11. "What is carbon black" (https://web.archive.org/web/20090401121937/http://www.carbon-black.org/what_is.html). International carbon black Association. Archived from the original (http://www.carbon-black.org/what_is.html) on 2009-04-01. Retrieved 2009-04-14.
12. Growney, David J.; Mykhaylyk, Oleksandr O.; Middlemiss, Laurence; Fielding, Lee A.; Derry, Matthew J.; Aragrag, Najib; Lamb, Gordon D.; Armes, Steven P. (2015-09-29). "Is Carbon Black a Suitable Model Colloidal Substrate for Diesel Soot?" (<https://doi.org/10.1021%2Facs.langmuir.5b02017>). *Langmuir*. **31** (38): 10358–10369. doi:10.1021/acs.langmuir.5b02017 (<https://doi.org/10.1021%2Facs.langmuir.5b02017>). ISSN 0743-7463 (<https://search.worldcat.org/issn/0743-7463>). PMID 26344920 (<https://pubmed.ncbi.nlm.nih.gov/26344920>). S2CID 206670654 (<https://api.semanticscholar.org/CorpusID:206670654>).
13. "Experimental and kinetic study of the interaction of a commercial soot toward NO at high temperature" (<http://www.combustion-institute.it/proceedings/MCS-7/papers/RKC/RKC-14.pdf>) (PDF). Archived (<https://web.archive.org/web/20130215105412/http://www.combustion-institute.it/proceedings/MCS-7/papers/RKC/RKC-14.pdf>) (PDF) from the original on 2013-02-15. Retrieved 2012-04-25.
14. Carbon Black - A Global Market Overview Jan 2016 • Industry Experts Report CP024 • 328 pages
15. Winter, J. and West FitzHugh, E., Pigments based on Carbon, in Berrie, B.H. Editor, Artists' Pigments, A Handbook of Their History and Characteristics, Volume 4, pp. 1–37.
16. Bone black (<http://colourlex.com/project/bone-black/>), ColourLex
17. Bomford D, Kirby J, Leighton, J., Roy A. Art in the Making: Impressionism. National Gallery Publications, London, 1990, pp. 112–119.
18. Édouard Manet, 'Music in the Tuileries Gardens' (<http://colourlex.com/project/manet-music-in-the-tuileries/>), ColourLex
19. Hennion, Marie-Claire (July 2000). "Graphitized carbons for solid-phase extraction". *Journal of Chromatography A*. **885** (1–2): 73–95. doi:10.1016/S0021-9673(00)00085-6 (<https://doi.org/10.1016%2FS0021-9673%2800%2900085-6>). PMID 10941668 (<https://pubmed.ncbi.nlm.nih.gov/10941668>).
20. Hu, Jingwei; Zhong, Shengwen; Yan, Tingting (2021-10-01). "Using carbon black to facilitate fast charging in lithium-ion batteries" (<https://www.sciencedirect.com/science/article/pii/S0378775321008545>). *Journal of Power Sources*. **508** 230342. Bibcode:2021JPS...50830342H (<https://ui.adsabs.harvard.edu/abs/2021JPS...50830342H>). doi:10.1016/j.jpowsour.2021.230342 (<https://doi.org/10.1016%2Fj.jpowsour.2021.230342>). ISSN 0378-7753 (<https://search.worldcat.org/issn/0378-7753>).

21. Younesi, Reza; Christiansen, Ane Sælland; Scipioni, Roberto; Ngo, Duc-The; Simonsen, Søren Bredmose; Edström, Kristina; Hjelm, Johan; Norby, Poul (2015). "Analysis of the Interphase on Carbon Black Formed in High Voltage Batteries" (<https://doi.org/10.1149%2F2.0761507jes>). *Journal of the Electrochemical Society*. **162** (7): A1289 – A1296. doi:10.1149/2.0761507jes (<https://doi.org/10.1149%2F2.0761507jes>). ISSN 0013-4651 (<http://search.worldcat.org/issn/0013-4651>). S2CID 53486824 (<https://api.semanticscholar.org/CorpusID:53486824>).
22. Dominko, Robert; Gaberscek, Miran; Drofenik, Jernej; Bele, Marjan; Pejovnik, Stane; Jamnik, Janko (2003-06-01). "The role of carbon black distribution in cathodes for Li ion batteries" (<https://www.sciencedirect.com/science/article/pii/S0378775303002507>). *Journal of Power Sources*. Selected papers presented at the 11th International Meeting on Lithium Batteries. 119–121: 770–773. Bibcode:2003JPS...119..770D (<https://ui.adsabs.harvard.edu/abs/2003JPS...119..770D>). doi:10.1016/S0378-7753(03)00250-7 (<https://doi.org/10.1016%2FS0378-7753%2803%2900250-7>). ISSN 0378-7753 (<https://search.worldcat.org/issn/0378-7753>).
23. Sorahan T, Harrington JM (2007). "A "lugged" analysis of lung cancer risks in UK carbon black production workers, 1951–2004". *Am J Ind Med*. **50** (8): 555–564. doi:10.1002/ajim.20481 (<https://doi.org/10.1002%2Fajim.20481>). PMID 17516558 (<https://pubmed.ncbi.nlm.nih.gov/17516558>).
24. Ward EM, Schulte PA, Straif K, Hopf NB, Caldwell JC, Carreón T, DeMarini DM, Fowler BA, Goldstein BD, Hemminki K, Hines CJ, Pursiainen KH, Kuempel E, Lewtas J, Lunn RM, Lynge E, McElvenny DM, Muhle H, Nakajima T, Robertson LW, Rothman N, Ruder AM, Schubauer-Berigan MK, Siemiatycki J, Silverman D, Smith MT, Sorahan T, Steenland K, Stevens RG, Vineis P, Zahm SH, Zeise L, Coglianò VJ (2010). "Research recommendations for selected IARC-classified agents" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2957912>). *Environmental Health Perspectives*. **118** (10): 1355–62. doi:10.1289/ehp.0901828 (<https://doi.org/10.1289%2Fehp.0901828>). PMC 2957912 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2957912>). PMID 20562050 (<https://pubmed.ncbi.nlm.nih.gov/20562050>).
25. Morfeld P, McCunney RJ (2007). "Carbon black and lung cancer: Testing a new exposure metric in a German cohort". *Am J Ind Med*. **50** (8): 565–567. doi:10.1002/ajim.20491 (<https://doi.org/10.1002%2Fajim.20491>). PMID 17620319 (<https://pubmed.ncbi.nlm.nih.gov/17620319>).
26. "Occupational Safety and Health Guidelines for carbon black: Potential Human Carcinogen, Centres of Disease Control and Prevention, National Institute for Occupational Safety and Health" (<https://www.cdc.gov/niosh/docs/81-123/pdfs/0102.pdf>) (PDF). Retrieved 2013-01-14.
27. "Occupational Safety and Health Guideline for Carbon Black: Potential Human Carcinogen" (<https://www.cdc.gov/niosh/docs/81-123/pdfs/0102.pdf>) (PDF). Centers of Disease Control and Prevention, National Institute for Occupational Safety and Health. Retrieved 11 January 2013.
28. "CDC – NIOSH Pocket Guide to Chemical Hazards – Carbon black" (<https://www.cdc.gov/niosh/npg/npgd0102.html>). www.cdc.gov. Retrieved 2015-11-27.

Further reading

- Doerner, Max. *The Materials of the Artist and Their Use in Painting: With Notes on the Techniques of the Old Masters*, Revised Edition. Harcourt (1984). ISBN 0-15-657716-X. This is a contemporary English language edition of a work originally published in German.
- Meyer, Ralph. *The Artist's Handbook of Materials and Techniques*. Fifth Edition, Revised and Updated. Viking (1991) ISBN 0-670-83701-6
- Carbon Black: A users guide (<https://web.archive.org/web/20070820050437/http://www.carb>

[on-black.org/user_guide.html](https://carbon-black.org/user_guide.html)). Published by the [International Carbon Black Association](https://carbon-black.org/).

External links

- [International Chemical Safety Card 0471 \(https://www.ilo.org/dyn/icsc/showcard.display?p_lang=en&p_card_id=0471&p_version=2\)](https://www.ilo.org/dyn/icsc/showcard.display?p_lang=en&p_card_id=0471&p_version=2)
 - ["Carbon black" \(https://www.cdc.gov/niosh/npg/npgd0102.html\)](https://www.cdc.gov/niosh/npg/npgd0102.html) – *NIOSH Pocket Guide to Chemical Hazards*, CDC website entry
 - ["Carbon Black Industry" \(https://tshaonline.org/handbook/online/articles/doc01\)](https://tshaonline.org/handbook/online/articles/doc01) from the *Handbook of Texas Online*
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